

Trains

In 1804 British engineer Richard Trevithick tried attaching a steam engine to a wagon, a job that before then had been done by horses. It easily pulled enormous weights, and the steam railway was born. Today's trains use diesel, electricity, or magnetic levitation to run fast and cleanly.

STEAM POWER

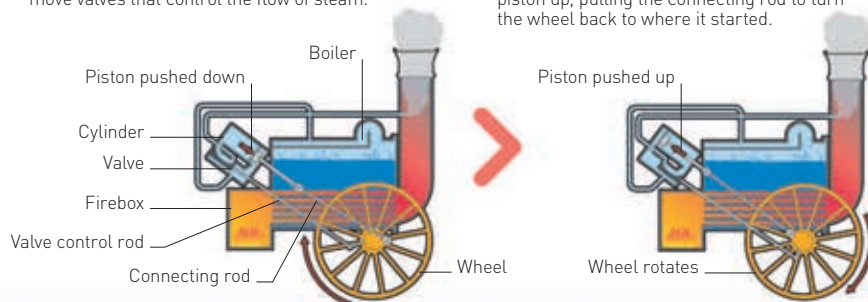
A steam engine runs on the heat energy that is produced by burning coal or other fuel. Inside the steam engine there is a fire that heats a boiler filled with water. The steam that is produced goes into cylinders and pushes pistons backwards and forwards. The pistons are connected to the driving wheels, and push and pull them round.

1 PISTON MOVES DOWN

Steam moves from the boiler into cylinders. The steam pressure pushes pistons, which move rods to turn the wheels. The turning wheels move valves that control the flow of steam.

2 PISTON MOVES UP

When the piston nears the bottom of the cylinder, the valve swaps the steam flow from the top to the bottom. The steam pushes the piston up, pulling the connecting rod to turn the wheel back to where it started.



DIESEL TRAINS

Steam engines polluted the air and were inefficient, so people began to look for better ways of powering trains. In 1892 the German engineer Rudolf Diesel invented the diesel engine that ran on a liquid type of fuel.



BOXLEY WHITCOMB 30-DM-31
1941

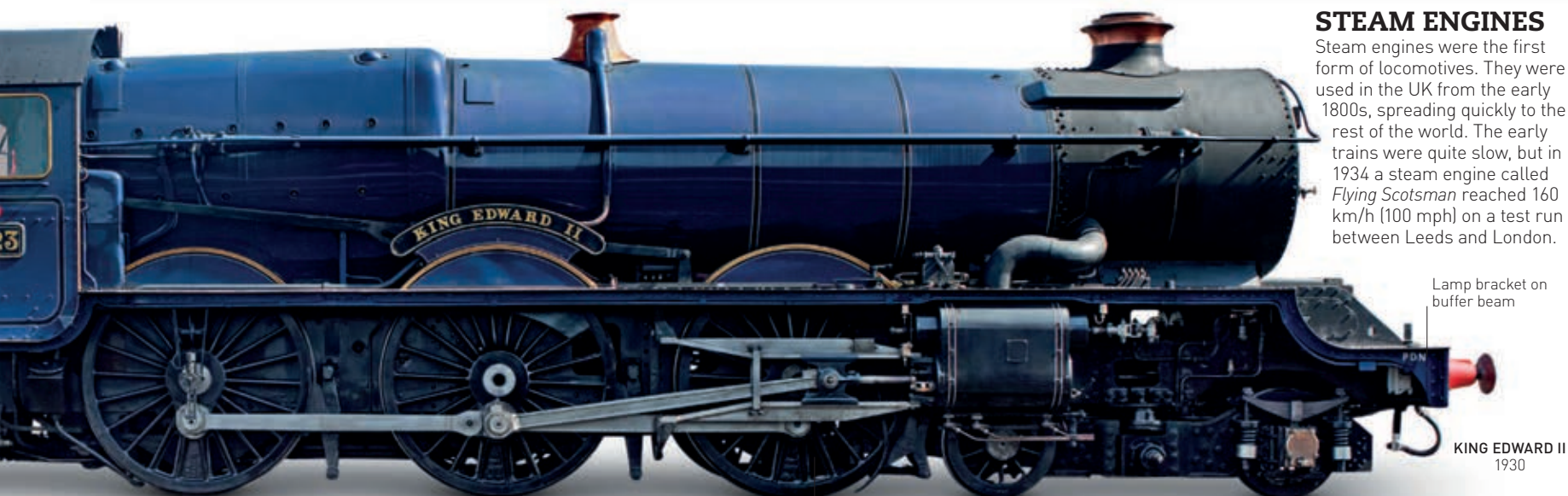


ENGLISH ELECTRIC DELTIC
1955

ONE EARLY DIESEL
PASSENGER TRAIN WAS
CALLED THE FLYING
HAMBURGER

STEAM ENGINES

Steam engines were the first form of locomotives. They were used in the UK from the early 1800s, spreading quickly to the rest of the world. The early trains were quite slow, but in 1934 a steam engine called *Flying Scotsman* reached 160 km/h (100 mph) on a test run between Leeds and London.



Lamp bracket on
buffer beam

KING EDWARD II
1930

Wheels driven by pistons



ROCKET
1829



TALIESIN
1876



MERDDIN EMRYS
1879



LB&SCR B1 CLASS
1882



LYR WREN
1887



NER (CLASS XL NO.66)
1902



VGN (CLASS SA NO.4)
1910



GWR HALL CLASS
1928



MALLARD
1938



HUNSLET AUSTERITY
1944



DR (CLASS 99.73-76)
1954